

The problem families

Outline

How to read these notes

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Introduction: the problem families

- ▶ **Twenty-three problems**, organized in families
- ▶ Each one applies, one at a time, the techniques of the modelling notes:
 - ▶ the **links between variables**
 - ▶ the **bounds from both sides**
 - ▶ the **protocol of the sandwich**
- ▶ All problems follow the same scheme: we see it once here

What you need to know first

- ▶ A reader arriving here without having read the modelling notes still finds everything needed in the **statement** and in the **model**
- ▶ The **proofs of the links**, however, are there
- ▶ That is the step that tells a **correct** model from a merely **plausible** one

What every exercise produces (1/3)

1. A **feasible and integer** solution from a constructive heuristic
 - ▶ checked on constraints, bounds *and* integrality
 - ▶ the integrality check is what makes the value a bound of the MILP and not of the relaxation alone
2. The **dual of the relaxation without the bounds**
 - ▶ written in general form and for the instance
 - ▶ with the directions taken from the conversion table

What every exercise produces (2/3)

3. A **feasible dual solution built by hand**

- ▶ with the recipe declared
- ▶ checked feasible and evaluated

4. The two **relaxations** $z(LP)$ and $z(LP^+)$ from the solver

- ▶ with a note on whether they coincide and why

What every exercise produces (3/3)

5. The **optimum** $z(MILP)$ and the table

$$UB \cdot LB \cdot z(LP) \cdot z(LP^+) \cdot z(MILP) \cdot \text{gap}$$

with the gap of the heuristic

6. The **additional considerations**:

- ▶ valid inequalities with their count
- ▶ constraints that preserve optimality, declared as such
- ▶ comparison of the relaxations of alternative formulations

Where the numbers come from

- ▶ Every number in the table exists in a **CSV** produced by the problem's script
- ▶ An assert in `check_numbers.py` compares it with the value quoted in the text
- ▶ So the numbers of the slides and of the notes are **reproducible**: it is enough to run the script again

Summary

The problem families:

- ▶ 23 problems applying links between variables, bounds from both sides and the protocol of the sandwich

The scheme of every exercise:

- ▶ primal bound UB from a heuristic (integer solution, checked)
- ▶ dual bound LB from a dual solution built by hand
- ▶ $z(LP)$, $z(LP^+)$, $z(MILP)$ from the solver, and the gap
- ▶ additional considerations on valid inequalities and formulations